



PIONEER JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS 2011

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H1 BIOLOGY

8875 / 02

PAPER 2

13 September 2011

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Read the instructions on the answer sheet
very carefully.**

This document consists of 2 sections:

Section A

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B

Answer Question 4.

Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 9
2	/ 14
3	/ 10
4	/ 7
Section B	
5 or 6	/ 20
TOTAL	/ 60

This document consists of **17** printed pages (inclusive of this page)

[Turn Over

Section A: Structured Questions [40 marks]

Question 1

An experiment was carried out to determine what happens to amino acids after they are absorbed by the animal cells. The cells were incubated in a medium containing radioactively labelled amino acid for 5 mins. The radioactive amino acids were then washed off and the cells were incubated in a medium containing only non radioactive amino acids.

Samples of the cells were removed from the cells every 5 mins for 40 mins. For each sample, the level of radioactivity in three different organelle A, B and C were determined.

The results of the experiment are shown in the graph below.

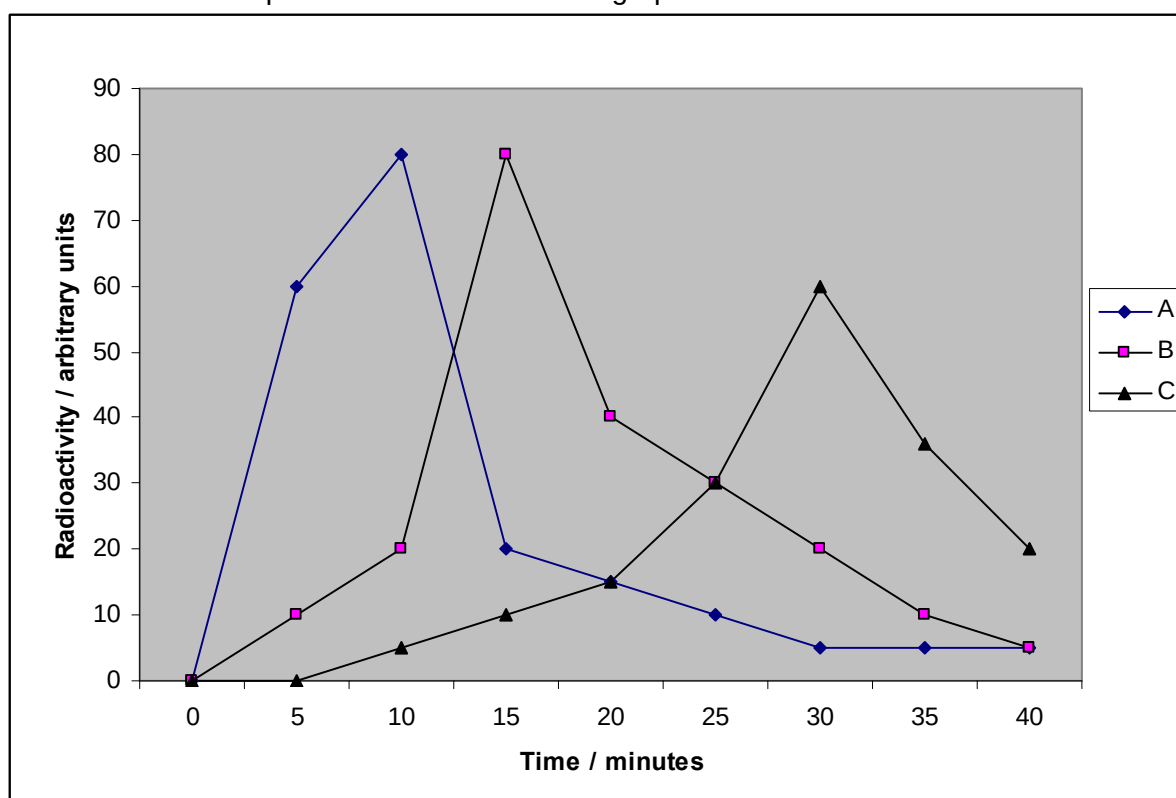


Fig 1.1

(a) Identify the organelle A, B and C by choosing from the list below. Write your answer in the space provided. [3]

Golgi apparatus; nucleus; mitochondria; ribosome; rough endoplasmic reticulum

Organelle A	ribosome
Organelle B	RER
Organelle C	golgi apparatus

(b) Explain briefly, with reference to the functions of the organelles, the changes in the levels of radioactivity of the three organelles with time. [4]

- **Increased in radioactivity (0arbitrary units to 80 arbitrary units) is detected in the first 10 mins in A(ribosome)**
- **which is the site for protein synthesis as**
- **radioactive amino acids are transported there to form polypeptide/protein**
- **Radioactive protein formed then enter B , RER for temporary storage**
- **As shown by increase in activity(20ar to 80ar) in B from 10-17 mins**
- **And a corresponding decrease in radioactivity(80ar to 18ar) in A**
- **Thereafter, radioactive proteins are transported to C golgi apparatus for chemical modification**
- **As shown by increase in activity(15ar to 60ar) in C from 20-30 mins and a corresponding decrease in radioactivity(20ar – 5ar) in B before being secreted out of C and out of the cell from 30-40mins**
- **As shown by decrease in radioactivity(60ar – 20ar)**

(c) Suggest what will happen to the level of radioactivity after 40 mins and explain your suggestion. [2]

- **Radioactivity will continue to drop in all organelle**
- **Referenc to half life(isotope)/ natural drop in radioactivity over time**
- **Radioactive proteins moved out of cell/exocytosis/secreted**
- **Radioactivity will not reach 0ar**
- **because some amino acid/ proteins are incorporated within the cell**

[Total: 9 marks]

Question 2

Ras gene is an oncogene that codes for a G-protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases. The cellular response at the end of the pathway is the synthesis of a protein that stimulates the cell cycle.

Mutations in *Ras* gene could lead to the development of cancer. The organization of the normal **Ras gene** is shown in **Figure 2.1**.

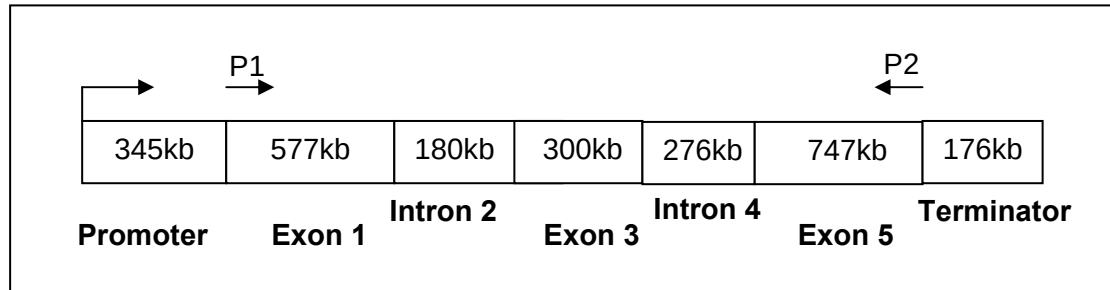


Fig 2.1

Using mRNA from in seven cell types in a cancer patient, reverse transcription was carried out. Subsequently, PCR was performed on the products of reverse transcription using primers P1 and P2, which are specific to the exon regions as shown in Figure 2.1. The PCR products were separated on an agarose gel and the results are shown in the gel image as seen in Figure 2.2.

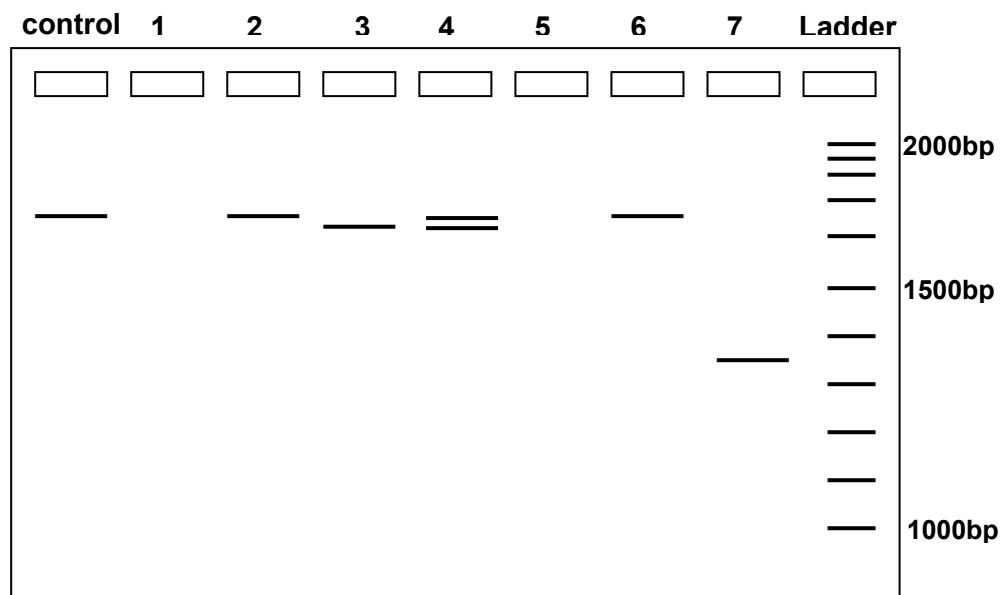


Fig 2.2

Figure 2.3 shows an analysis of Ras protein in the seven cell types in a cancer patient. The control shows the amount of the two substances in a normal cell. A darkly stained band indicates that there is high amount of the respective substance in that particular cell type.

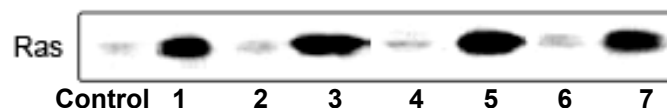


Fig 2.3

(a) Explain how the DNA fragments can be separated by electrophoresis. [2]

- **The negatively charged DNA(due to the phosphate groups) will migrate towards the positively charged pole / anode.**
- **The agarose gel forms a cross-linked matrix that functions as a 'molecular sieve'.**
- **DNA fragments are separated by size.**
- **smaller DNA fragments have lower resistance in moving through the pores of the gel → move further from the wells / the larger fragments move slower due to higher resistance → are closer to the wells**

(b) With reference to the gel analysis seen in Figure 2.2 and 2.3, explain how cell type 1 and cell type 5 could become cancerous. [3]

- **mutation could have occurred at P1 of exon 1 and/or P2 of exon 5 regions.**
- **the mutation thus caused the sequence on the DNA to be not complementary to the primers**
- **→ prevents the formation of hydrogen bonds between the complementary bases.**
- **mutation led to the formation of over expression of the Ras proto-oncogene**
- **= thick band of ras protein**
- **due to gain of function mutation**

(c) Cell type 4 was confirmed as a cancerous cell line. With reference to the gel analysis seen in Figure 2.2 and 2.3, suggest an explanation on how cell type 4 could become cancerous. [3]

- **dominant mutation**
- **one copy of a normal ras gene as seen in the band nearer to the well**
- **the other copy is a mutated ras gene with base deletion → band further away from the well indicating it is a smaller DNA fragment.(between 1600 to 1623 bases.)**
- **however, normal levels of ras protein of 1624bases detected**
- **could imply the production of overactive ras protein**
- **gain of function mutation**

(d) Figure 2.4 shows the steps involved in the polymerase chain reaction used to obtain the results as seen in Figure 2.2.

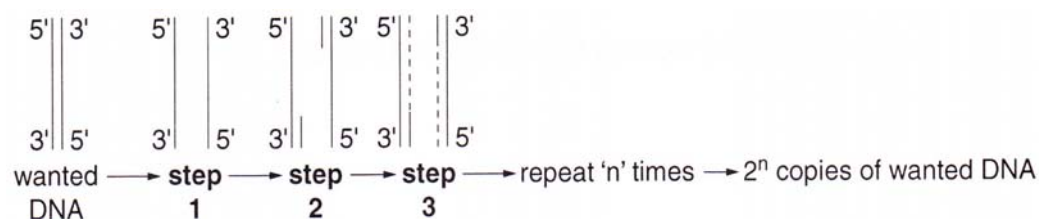


Figure 2.4

With reference to Fig 2.4, describe each of the three steps of one cycle of the PCR.

Step 1 [2]:

- strands are separated by heating to 95°C
- for two minutes.
- This breaks the hydrogen bonds.
- Hence, the DNA double helix can now unwind and separate into two separate single strands to allow for DNA polymerase to start replicating each strand.

Step 2 [2]:

- PCR requires short lengths of DNA (about 20 bp long) called primers, to get it started.
 - They are synthesized separately and have to be added for DNA polymerization to occur.
- The DNA must be cooled to 50-65°C to allow the primers to anneal to
- 3' end of their complementary sequences on the separated DNA strands.
- Cooling allows primers to bind by hydrogen bonding to the ends of the target sequence, one primer on each strand.
- Because primers are in excess and are short sequences, they will anneal to their long target sequences before the two original strands can come back together.

Step 3 [2]

- The mixture is heated to 72°C which is the optimum temperature for the heat stable *Taq* polymerase
- The *Taq* polymerase will extend the primers and complete replication of the rest of the DNA.
- It does this by adding nucleotides to the 3' ends of the primers using the longer DNA strands as templates/extends new DNA strands in the 5' to 3' direction.
- It catalyses the formation of phosphodiester bonds between adjacent nucleotides as the new DNA strand elongates.
- Hence at the end of this extension stage, hydrogen bonds between newly added nucleotides and template strands as well as phosphodiester bonds between newly added nucleotides will hold the 2 DNA molecules together until the next cycle starts over again.

[Total: 13 marks]

Question 3

Figure 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

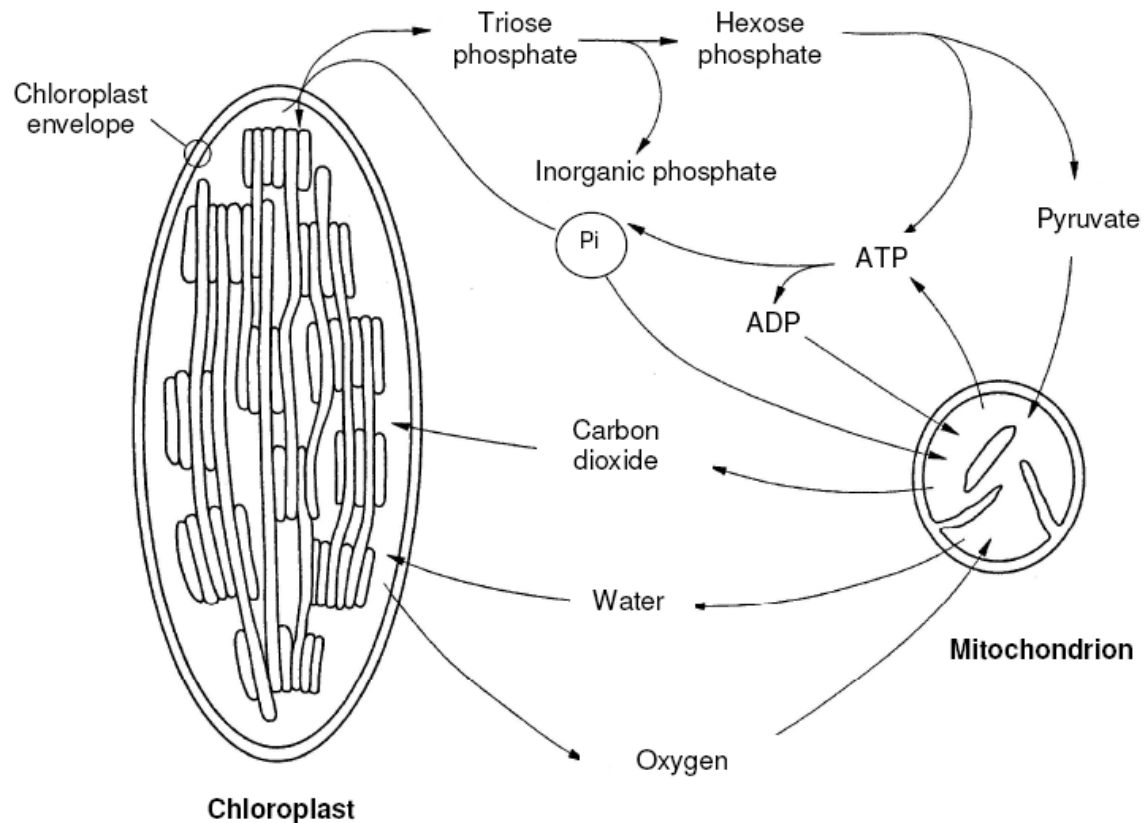


Fig. 3.1

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Figure 3.1**, outline how the following will link the metabolic pathways between the chloroplast and mitochondrion of this cell:

triose phosphate and hexose phosphate [3],

- In the stroma; of the chloroplast,
- triose phosphate is formed as a product of the light-independent/dark reaction;
- Triose phosphate moves out of the chloroplast through carrier proteins present in the chloroplast envelope;
- Triose phosphate is then converted to hexose phosphate that can enter glycolysis;
- as an respiratory substrate/intermediate;
- to be eventually completely oxidized; in aerobic respiration; /
- hexose phosphate is broken down to pyruvate;

- during glycolysis; in the cytosol.
- Pyruvate then enters the mitochondrial matrix to undergo the link reaction;

(b) With reference to **Figure 3.1**,

(i) explain how it is possible for the oxygen concentration to remain constant. [2]

- Oxygen released by chloroplast during the light dependent reaction of photosynthesis/photolysis of water;
- is used by the mitochondrion as the final electron acceptor in the ETC of cellular respiration;
- Compensation point has been reached;
- whereby the rate of respiration is equal to the rate of photosynthesis;
- hence no net change in oxygen concentration;

(ii) explain why there is no build up in the concentration of phosphate ions inside the mitochondria as a result of the inward passage of phosphate ions. [2]

- Phosphate ions are used to produce ATP;
- in oxidative phosphorylation/Kreb cycle/chemiosmosis/electron transport;
- of cellular respiration;
- These ATP subsequently leaves the mitochondria;
- Phosphate ions can also be used in substrate level phosphorylation in the Krebs Cycle;
- hence phosphate ions do not accumulate in the mitochondria.

An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell. **Figure 3.2** below shows the various experimental systems that are prepared in an attempt to synthesize ATP.

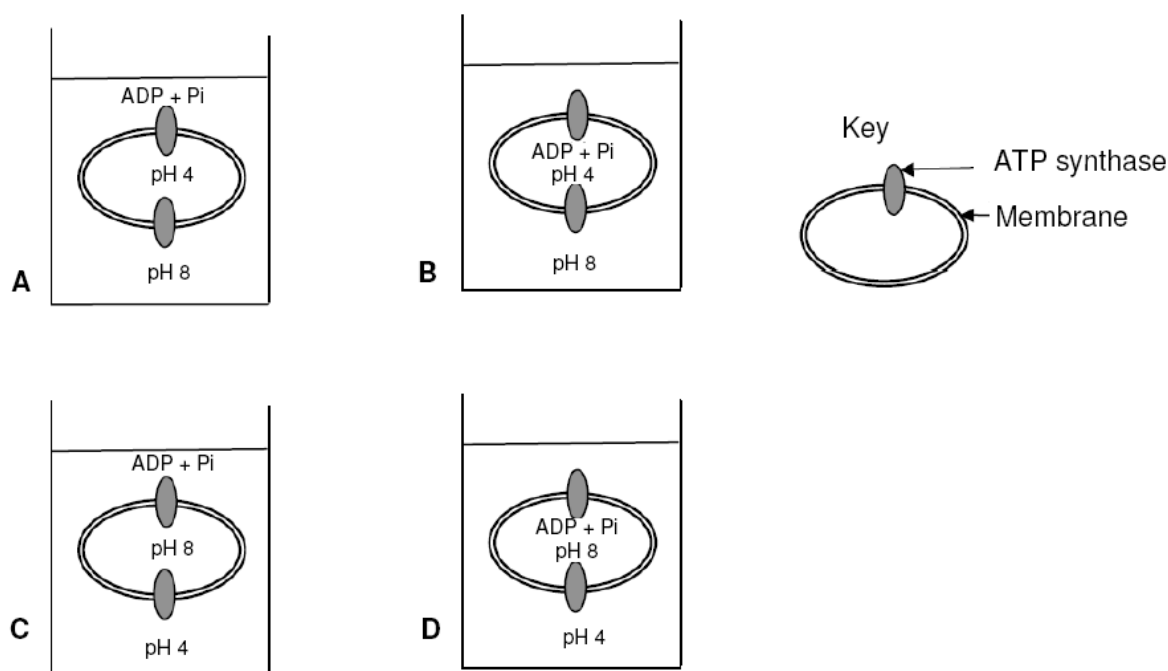


Figure 3.2

(c) Which experimental system (A to D) shown in **Figure 3.2** would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer. [3]

- **Experimental system A;**
- The pH within the membrane is pH 4 compared to pH 8 outside the membrane. This means that the concentration of hydrogen ions is higher within the membrane than outside the membrane (i.e. lower concentration of hydrogen ions).
- This generates a proton motive force / proton gradient / electrochemical gradient;
- Hence, there is a tendency for the hydrogen ions to leave the interior of the vesicle; via the ATP synthase;
- down their concentration gradient;
- when hydrogen ions flows out of the membrane. ATP is formed using the ADP and Pi found outside the membrane;
- (Reject: Combination of answers e.g. Experimental systems A and D)

[Total: 10 marks]

Question 4

Cytochrome c is found in mitochondria of every aerobic eukaryote – animal, plant and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. These molecules are homologous.

Table 4.1 shows the first 16 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath.

A ditty (“) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

Cytochrome c	Amino acid																
		1			4				8				12				16
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln
Pig		“	“	“	“	“	“	“	“	“	“	Val	Gln	“	“	Ala	“
Chicken		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Turkey		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Dogfish		“	“	“	“	“	“	“	“	Val	“	Val	Gln	“	“	Ala	“
<i>Drosophila</i>	<<<	“	“	“	“	“	“	“	“	Leu	“	Val	Gln	Arg	“	Ala	“
Wheat	<<<	“	Asn	Pro	Asp	Ala	“	Ala	“	“	“	Lys	Thr	“	“	Ala	“
Yeast	<<<	“	Ser	Ala	Lys	“	“	Ala	Thr	Leu	“	Lys	Thr	Arg	“	“	“

Table 4.1

(a) With reference to **Table 4.1**, explain what is meant by “These molecules are homologous” (line 7). [2]

- The cytochrome molecules are similar in that the different cytochromes carry out the same function but their sequences of amino acid in the polypeptide chain are not the same.
- The cytochrome molecule of the various organisms is a shared characteristics derived from a common ancestor;
- the cytochrome molecules share the same amino acids at many positions / are conserved;
- For all organisms, the molecules have the same amino acid at positions 1, 6, 10, 14, 16 from the N-terminal.

(b) Explain why the analysis of amino acid sequence can be used to establish evolutionary relationship. [3]

- **Amino acid sequence of a protein is coded by DNA nucleotide sequence.**
- **Mutations in the cytochrome gene resulted in differences in amino acid sequences;**
- **Mutations become fixed in a population / are inherited;**
- **Over time, as such each population accumulates its own set of mutations, the original DNA sequences diverges;**
- **The more similar the amino acid sequence from two species matches, the more likely gene coding for this protein evolved from a common gene**
- **from a shared ancestor.**

(c) State one limitation using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

- **One gene represents a small percentage of the genetic information in the organism**
- **OR**
- **is not representative of the entire genome.**

Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for chicken and turkeys, their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(d) Explain why different species of birds such as chicken and turkey can have the same amino acid sequence in their cytochrome c. [1]

- **Chickens and turkeys might have evolved from a common ancestor;**
- **fairly recently and so have the same amino acid sequences for cytochrome c.**

[Total: 7 marks]

Section B: Essay [20 marks]

Write your answer on the lines provided.

Question 5

- (a) Describe the structure of an amino acid and how a peptide bond is formed with another amino acid. [6]
- (b) Explain what is meant by primary secondary, tertiary and quaternary structure of a named protein. [8]
- (c) Outline how the structure of a named globular protein is related to its specific function. [6]

OR

Question 6

- (a) Describe the main features of homologous chromosomes. [6]
- (b) Explain how mitosis produces genetically identical cells. [6]
- (c) Explain the mode of action of enzymes. [8]

(a) Describe the structure of an amino acid and how a peptide bond is formed with another amino acid. [6]

- Each amino acid consists of a carbon atom (also known as the α -carbon) bonded covalently to four groups:
 1. a basic amino group ($-\text{NH}_2$),
 2. an acidic carboxyl group ($-\text{COOH}$),
 3. a hydrogen atom ($-\text{H}$), and
 4. a variable functional group, the R group (also known as the side chain)
- ✧ Amino acids are joined together by a type of bond known as a *peptide bond*.
- ✧ A peptide bond is formed when a water molecule is eliminated
- ✧ during a condensation reaction
- ✧ between the amino group of one amino acid and the carboxyl group of another.
- ✧ It has a free amino ($-\text{NH}_2$) group at one end (known as N-terminal or amino terminal), and a free carboxyl ($-\text{COOH}$) group at the other end (known as C-terminal or carboxyl terminal)
- ✧ process is catalysed an enzyme found in the ribosomes
- ✧ when the amino acids are attached to aminoacyl tRNA complexes at the A and P sites
- ✧ during the process of translation.

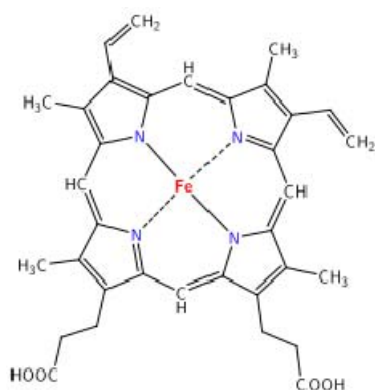
(b) Explain what is meant by primary secondary, tertiary and quaternary structure of a named protein. [8]

- A protein with all 4 levels of structure is that of a globular protein, haemoglobin.
- Primary structure refers to the number, type and unique sequence of amino acids found

- in each of the 4 polypeptide chains (2 alpha and 2 beta chains) linked by peptide bonds in a polypeptide chain. –
- synthesized by ribosome
- Secondary structure: segments of their polypeptide chain repeatedly coiled or folded
- into some kind of geometrically regular secondary structure.
- The 2 main kinds are the α -helix and the β -pleated sheet.
- Both structures are stabilized by hydrogen bonds between the amino and carboxyl groups of two different amino acids.
- tertiary structure is the overall compact, globular, 3-D structure of each haemoglobin polypeptide chain
- conformation of a polypeptide resulting from interactions between side-chains of the various amino acids in a polypeptide chain.
- The compact structure is maintained by four types of bonds, namely *hydrogen bonds*, *ionic bonds*, *hydrophobic interactions* as well as the stronger *disulphide bonds*.
- As the polypeptide folds into its functional conformation, amino acids with hydrophobic (non-polar) side chains congregate in clusters via hydrophobic interactions at the core of the protein, out of contact with water.
- Meanwhile, hydrogen bonds between polar side-chains and ionic bonds between positively and negatively charged side-chains also help stabilize tertiary structure.
- These are all weak interactions but their cumulative effect help to stabilize and give the protein a unique shape.
- Quaternary structure refers to the specific orientation of 2 or more polypeptide chains with respect to one another and the nature of interactions that stabilize this orientation.
- Each polypeptide chain in such a protein is called a subunit
- There are 4 subunits for haemoglobin, namely 2 alpha and 2 beta polypeptide chains
- The subunits are held together by hydrogen bonds, ionic bonds and hydrophobic interactions to form a multimeric functional protein.

(c) Outline how the structure of a named globular protein is related to its specific function. [6]

Haemoglobin – globular protein with physiological and transport role



The iron ion may either be in the Fe^{2+} or Fe^{3+} state, but ferrihemoglobin ([methemoglobin](#)) (Fe^{3+}) cannot bind oxygen.^[22] In binding, oxygen temporarily oxidizes (Fe^{2+}) to (Fe^{3+}), so iron must exist in the +2 oxidation state to bind oxygen

Characteristics	Structure	Function
Unique 3D conformation	Alpha & beta globin chains folded into unique compact 3D globular shape = tertiary structure	compact for packing more Hb into rbc, hence carriage & transport of more O_2 and CO_2 ;
Conjugated protein	Conjugated protein → Each polypeptide chain accommodates a prosthetic haem group with ferrous ion / iron atom of haem group,	capable of binding oxygen molecules → carriage of oxygen molecules to respiring tissue & CO_2 to lungs for excretion;
Role of Fe^{2+}	Fe^{2+} within each haem group allows for the reversible binding of oxygen molecule and CO_2 (because bond is weak)	oxygen molecule readily released to respiring tissue or CO_2 readily released in lungs for excretion;
No. of polypeptide chain making up quaternary structure	Quaternary structure - 4 globin chains & thus 4 haem groups;	Increase no. of oxygen molecules bound to Hb → 4 Oxygen molecules per haemoglobin; More efficient transport of oxygen molecules to respiring tissues
Solubility in water	Globin chains folded with hydrophobic R side chains of the amino acids pointing inwards/core of haemoglobin and hydrophilic R side chains of the amino acids facing outwards into aqueous	Outwardly pointing hydrophilic R groups → Important for Hb to be soluble in water → for mobility around body via the bloodstream to pick up oxygen and carbon dioxide that have been

	environment	dissolved in the blood.
hydrophobic R groups	point inwards to the center of the molecule and its hydrophilic R groups face outwards, making it soluble in water.	The interactions between the hydrophobic R groups inside the molecules are important in holding it in its correct three-dimensional shape.
Quaternary structure of 4 subunits	Binding of 1 oxygen molecule to iron of one haem group will cause a change in the shape of the remaining 3 subunits, making it easier for the 3 to pick up oxygen or vice versa → Cooperative binding	Facilitates cooperative binding between 4 subunits → easier binding and release of O₂ - saturated in the lungs with high O₂ tension, rapidly dissociates over a small change in partial pressure of O₂, giving its oxygen to respiring tissue;

- As typical of globular proteins, each beta or alpha globin chain has
- The outward-pointing hydrophilic R groups on the surface of the molecules are important in maintaining its solubility in the aqueous environment.

Question 6

- (d) Describe the main features of homologous chromosomes. [6]
(e) Explain how mitosis produces genetically identical cells. [6]
(f) Explain the mode of action of enzymes. [8]

Describe the main features of homologous chromosomes. [6]

- **Matching pair of chromosomes,**
- **one from maternal and the other from paternal origin.**
- **They have same structural features eg: same gene loci (thus same sequence of genes),**
- **same number of genes**
- **thus same length and**
- **same centromere position.**
- **Same no. of genes arranged in the same order**
- **They are genetically similar but not identical as different alleles might be present.**
- **After semi-conservative DNA replication, each homologous chromosomes will exist in the form of a double structure →**
- **formation of 2 identical sister chromatids.**
- **Homologous chromosomes pair up during prophase I of meiosis as a bivalent. This enables crossing over of non-sister chromatids to take place. → creates genetic variation.**

Explain how mitosis produces genetically identical cells. [6]

- **Process by which a nucleus divides to produce two daughter nuclei**
- **Each containing the same number of chromosomes as the original nucleus.**
- **During the S phase, DNA replicates, producing two identical molecules of DNA which forms identical sister chromatids.**
- **This doubling of DNA content ensures that after separating, each cell receives the same amount of DNA as the parent cell, so as to produce genetically identical cells.**
- **The sister chromatids is genetically identical to each other and thus have the same base sequence as well as alleles.**
- **During metaphase, chromosomes lined up along the equator place in a single file.**
- **The arrangement of the chromosomes on the spindle during metaphase ensures that the chromosomes are shared equally between the 2 daughter nuclei.**
- **During anaphase, the spindle fibres pull the sister chromatids apart. Centromeres divide.**
- **Upon separation, the chromatids are now considered as chromosomes.**
- **One copy of the chromosomes would be on each side of the pole.**
- **During telophase, The chromosomes reach their respective poles and become extended and indistinct.**
- **The nucleoli reappear and a nuclear envelope develops round each cluster of daughter chromosomes forming a nucleus.**
- **Each daughter nucleus now has one of the two identical chromatids of each chromosome of the parent nucleus. The daughter nuclei are thus genetically identical.**

Explain the mode of action of enzymes.

- **Enzymes are catalysts which speeds up a chemical reaction but remains unchanged at the end of reaction.**
- **Enzymes increase the rate of reaction but remain unaltered at the end of the reaction. Thus, they can be reused and are required only in small amounts.**
- **Enzymes possess active sites where reaction takes place. The active sites have a specific 3D conformation into which only certain specific substrate molecules will fit.**
- **This specificity explains why most enzymes can only work on one particular type of substrate or a group of substrates.**
- **Enzyme activity is affected by changes in pH, temperature, substrate and enzyme concentration and the presence of cofactors and inhibitors.**
- **Most enzymes are globular proteins and are far larger than the substrates they act on. The 3-dimensional conformation is due to interactions between amino groups, carboxyl groups and R groups via intramolecular hydrogen bonds, ionic interactions, disulphide bonds or hydrophobic interactions. The catalytic activity of an enzyme depends on the its protein conformation.**
- **Collision between substrate and enzyme at the correct orientation causes the substrate molecules to bind to the enzyme molecule at its active site to form an enzyme-substrate (E-S) complex.**
- **Enzymes and substrate are usually held together by weak bonds, such as ionic, hydrogen and hydrophobic bonds, which can be made and broken rapidly.**
- **Enzymes increase rate of reaction by lowering the activation energy via**

- active site held in precise orientation – reactants in close proximity and in correct orientations for rxn to take place;
- active site create physical stress in substrate - bonds are strained and hence break more easily to form new bonds in product;
- active site changes reactivity of substrate – R-groups of aa at active site may increase reactivity of active site by changing charge of substrate.
- at the end of the reaction the products leave the active site.
- Enzymes are able to catalyse a reaction as they have a region known as active site
- Substrate fit into the enzyme via induced fit hypothesis (no mark)
- As the substrate enters the active site, the amino acids, which make up the active site are moulded or 'induced' into a precise shape complementary to the substrate.
- An enzyme-substrate complex is formed.
- Induced fit brings a greater interaction between the chemical groups of the substrate and active site/ alignment of catalytic groups of enzymes for reaction to take place more effectively.
- At the end of the reaction, products leave the active site
- and the enzyme returns to its normal shape.

~THE END~